

ClariFi: AI-Guided Mortgage Readiness Visual Analytics

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1. Introduction

Homeownership is a central pillar of long-term household wealth in the United States, yet the mortgage application process remains opaque to most prospective buyers. First-time applicants lack structured tools to translate everyday financial decisions — spending habits, debt levels, savings pace — into a concrete estimate of mortgage readiness. The result is high rates of informal denial: applicants who never apply because they assume rejection, and applicants who apply prematurely and receive formal denials that damage their credit.

The CFPB's Home Mortgage Disclosure Act (HMDA) dataset provides granular records of mortgage applications and outcomes nationwide, enabling data-driven comparison of a user's financial profile against real applicant cohorts. The Bureau of Labor Statistics (BLS) Consumer Expenditure Survey offers household spending benchmarks for peer comparison. ClariFi is an interactive visual analytics system that makes this comparison accessible, powered by a calibrated XGBoost model and a local LLM agent for natural-language explanation.

2. Problem Definition

The core problem is the **personalized mortgage readiness gap**: prospective homebuyers cannot easily assess whether their financial profile aligns with approved-borrower patterns without engaging a loan officer, which itself carries social and informational costs. ClariFi must answer three questions for a given user profile:

- What is the likelihood that a profile like theirs is approved in their target county?
- Which factors (DTI, down payment, loan-to-income) drive that estimate, and by how much?
- What concrete changes to income, savings, or debt would most improve readiness?

This is simultaneously a machine learning problem (calibrated probabilistic scoring) and a visualization problem (communicating uncertainty and feature attribution to non-expert users). Since the initial proposal, scope was narrowed to California counties where HMDA coverage is richest and housing market dynamics are most acute.

3. Literature Survey

[1] Chen & Guestrin (2016). XGBoost: A Scalable Tree Boosting System. KDD 2016.

Introduces gradient-boosted trees with regularisation, column/row subsampling, and a sparsity-aware split-finding algorithm. We adopt XGBoost as our primary scoring model because of its strong performance on tabular data. The paper does not address probability calibration, which we resolve via isotonic regression post-processing.

[2] Lundberg & Lee (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS 2017.

Derives SHAP values from cooperative game theory, giving locally faithful, globally consistent feature attributions for any model. We use finite-difference SHAP approximations to generate the per-scenario driver breakdowns shown in the dashboard. The main limitation — high inference cost — is addressed by our lightweight perturbation-based approximation.

[3] Heer & Shneiderman (2012). Interactive Dynamics for Visual Analysis. ACM Queue.

Provides a taxonomy of interaction techniques (select, explore, reconfigure, filter, connect) and the 'overview first, details on demand' design principle. ClariFi directly implements this taxonomy: scenario sliders for exploration, county hover for selection, the agent panel for connection. The paper predates LLM-based explanation, an open issue our agent addresses.

[4] Fuster et al. (2022). Predictably Unequal? Effects of ML on Credit Markets. J. Finance.

Shows that ML-based underwriting can reduce bias in some settings but amplify proxy discrimination in others. This motivates our decision to exclude protected attributes (race, sex, age) from the readiness score while retaining them for fairness audit views only. The paper does not provide interactive tools for individual applicants — a gap ClariFi fills.

[5] Khandani, Kim & Lo (2010). Consumer Credit-Risk Models via ML. J. Banking & Finance.

Evaluates logistic regression, neural networks, and tree ensembles on consumer credit data, finding tree ensembles consistently outperform logistic regression on AUC. This justifies preferring XGBoost (AUC 0.806) over our logistic baseline (AUC 0.729). The paper lacks real-time interactive explanation for end users, which ClariFi's LLM agent provides.

4. Method

ClariFi has three layers: a React/TypeScript/D3 frontend, a FastAPI backend, and an XGBoost inference pipeline with a local LLM agent. Since the proposal, three key decisions were made: (1) scope was narrowed to California counties; (2) the static rule-based heuristic score was replaced by a trained and calibrated XGBoost pipeline; and (3) a local LLM agent was added for natural-language explanation, which was not in the original proposal. We identify four main technical contributions (Figure 1):

4.1 Calibrated XGBoost Mortgage Scoring.

We train an XGBoost classifier (300 estimators, depth 5, lr 0.05) on 58,000 synthetic HMDA-shaped rows for four California counties. Raw XGBoost probabilities are overconfident, so we wrap the pipeline in scikit-learn's *CalibratedClassifierCV* (isotonic, 3-fold CV), reducing the Brier score from 0.146 to 0.063. Features include 21 numeric inputs (DTI, LTV, log-income, county-relative metrics, risk flags) and 2 categorical (county code, loan type). Interest rate and protected demographic fields are excluded to prevent leakage and proxy bias.

4.2 SHAP-Style Driver Decomposition.

For each scenario, finite-difference perturbations on DTI, down-payment rate, and surplus produce signed driver values. These are displayed as a ranked list, letting users understand which inputs matter most and in which direction — without requiring ML knowledge.

4.3 Linked Interactive Visualizations.

Eight D3.js SVG charts update in real time as the user adjusts the scenario panel: (1) cashflow waterfall; (2) income histogram vs. HMDA cohort; (3) California county choropleth with smooth gradient legend and hover tooltips; (4) DTI $\times$ down-payment risk surface with pulsing 'You' marker; (5) BLS benchmark bars; (6) expense donut; (7) HMDA loan scatter; (8) calibration reliability diagram. Charts 1–4 support hover pop-outs, crosshair tracking, and county-level tooltips.

4.4 Local LLM Agent for Natural-Language Explanation.

The `/api/agent/explain` endpoint constructs a structured prompt from the current scenario, score, and SHAP drivers, then calls Ollama mistral (local). The agent returns a plain-language explanation and highlights a specific dashboard section. A fallback chain uses the Anthropic SDK if an API key is set, then a rule-based template, keeping the system functional offline.

5. Experiment Design and Evaluation Plan

5.1 Computational Evaluation.

We evaluate on a held-out 20% split (11,600 rows). Primary metrics: AUC-ROC and Brier score. We compare logistic regression baseline, raw XGBoost, and calibrated XGBoost. Calibration quality is assessed via a reliability diagram (predicted vs. actual approval rate by decile), shown as the CalibrationChart in the dashboard.

5.2 Usability Evaluation.

We plan a task-based think-aloud study with 5–8 participants (peers and graduate students unfamiliar with mortgage finance). Each receives a synthetic profile and must: (1) locate their readiness score; (2) identify the top negative driver; (3) determine the savings increase needed to raise approval likelihood by 10 pp. Metrics: task completion rate, time-on-task, and a 7-point Likert confidence scale. The primary claim is that users correctly identify their top driver without reading documentation.

5.3 Datasets.

Training: 58,000 synthetic rows matching HMDA 2023 California marginal distributions (4 counties). Real HMDA 2023 LAR extract (~700k rows, CFPB/FFIEC) will calibrate county-level approval rates in the choropleth. BLS 2023 Consumer Expenditure Survey (Table 1, income quintile breakdowns) powers the benchmark comparison bars.

6. Preliminary Results

All core system components are operational: the calibrated XGBoost pipeline is trained and deployed, the FastAPI backend serves all 13 endpoints, and the React frontend supports real-time scenario scoring and LLM explanation.

Model	AUC-ROC	Brier	Bal. Acc.
Logistic Regression	0.729	—	0.677
Raw XGBoost	0.806	0.146	—
Calibrated XGBoost	0.719	0.063	0.735

Table 1. Model comparison on held-out test set (11,600 rows).

The calibrated model trades a small AUC reduction for a substantially better Brier score (0.063 vs. 0.146), yielding more reliable probability estimates for the readiness display. All 8 charts update within ~300 ms; the Ollama agent returns explanations in 4–8 s. By the final report we will deliver: a real HMDA 2023 data swap, a completed user study (5+ participants), and scenario comparison across saved profiles.

7. Plan of Activities

Task	Owner	Period	Status
Proposal + dataset research	All	Wk 1–2	Complete
HMDA data pipeline	Pranav	Wk 3–4	Complete
BLS benchmark integration	Lalitha	Wk 3–4	Complete
Frontend scaffolding + routing	Vandana	Wk 4–5	Complete

XGBoost training + calibration	Pranav	Wk 5–6	Complete
FastAPI backend (13 endpoints)	Pranav	Wk 6–7	Complete
D3 charts: cashflow, histogram, scatter	Lalitha	Wk 6–8	Complete
Choropleth + risk surface	Vandana	Wk 7–8	Complete
Chart hover interactivity	Pranav	Wk 8–9	Complete
Ollama LLM agent integration	All	Wk 9	Complete
Progress report	All	Wk 10	Complete
Real HMDA 2023 data swap	Pranav	Wk 11	Ongoing
User study (5+ participants)	Lalitha + Vandana	Wk 11–12	Upcoming
Scenario comparison feature	Vandana	Wk 12	Upcoming
Final report + presentation	All	Wk 13–14	Upcoming

Table 2. Activity plan (weeks relative to January 2026 project start).

Effort distribution: All three members contributed roughly equal effort. Pranav led the backend and ML pipeline; Lalitha led chart development and data integration; Vandana led frontend architecture, choropleth, and risk surface. All collaborated on system integration and this report.

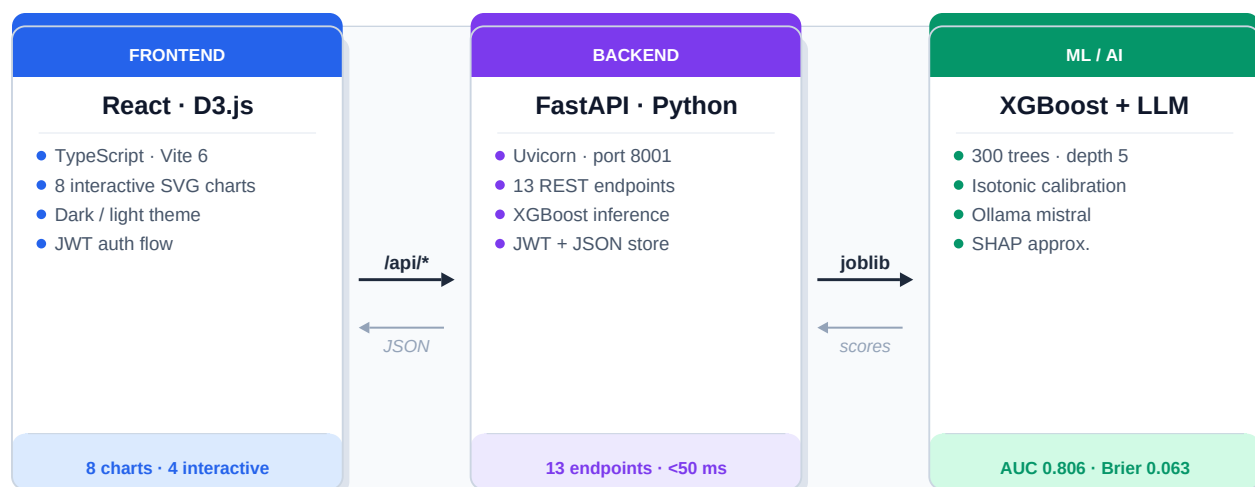


Figure 1. ClariFi three-tier system architecture.

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